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Summary

This paper introduces a framework designed to quantify the trade-offs between prediction accuracy and screening access within the context of policy evaluation, particularly when targeting the worst-off individuals in labor market settings. The authors propose a policy value function,

denoted as $V(\alpha, \beta, R^2)$, which is defined using the cumulative distribution function of a bivariate normal distribution. This function aims to capture the non-linear interplay between the screening threshold (α), the targeted outcome quantile (β), and the model's predictive performance (R^2). The core idea is to provide a metric that can guide policy decisions by evaluating the impact of changes in both screening thresholds and predictive accuracy. To further facilitate this evaluation, the authors introduce the Prediction-Access Ratio (PAR), which quantifies the relative gains achieved by increasing the screening threshold versus improving prediction accuracy. The paper's methodology involves deriving the policy value function under the assumption of a bivariate normal distribution for the joint distribution of covariates and outcomes, and then using this function to analyze the impact of changes in the key parameters. The empirical component of the paper consists of simulation experiments conducted on synthetic datasets. These experiments explore various scenarios, including random screening, near-perfect prediction, and capacity gap analysis, to demonstrate the potential of the proposed metrics. The authors also employ a residual scaling method to simulate improvements in prediction accuracy without retraining the model. The main findings from the experiments suggest that modest improvements in either prediction accuracy or screening capacity can lead to significant policy benefits. The paper concludes by suggesting that the framework can provide actionable insights for policymakers when allocating resources between improving predictive models and expanding screening capacity. Overall, the paper attempts to address a relevant problem in policy evaluation by providing a quantitative framework for balancing prediction accuracy and screening access, but it suffers from several limitations that impact the validity and generalizability of its findings.

✓ Strengths

The paper's primary strength lies in its attempt to address a relevant and complex problem in policy evaluation: the trade-off between prediction accuracy and screening access. This is a crucial consideration for policymakers who must often decide how to allocate limited resources between improving predictive models and expanding the reach of their programs. The introduction of the policy value function, $V(\alpha, \beta, R^2)$, and the Prediction-Access Ratio (PAR) represents a novel approach to quantifying this trade-off. By using the bivariate normal CDF, the authors attempt to capture the non-linear relationship between the screening threshold, the targeted outcome quantile, and the model's predictive performance. This provides a unified framework for evaluating policy interventions, which is a valuable contribution. The paper also attempts to provide empirical support for its framework through simulation experiments. These experiments explore various scenarios, including random screening, near-perfect prediction, and capacity gap analysis, which provide a comprehensive evaluation of the proposed metrics. The use of a residual scaling method to simulate improvements in prediction accuracy is also a practical approach that allows the authors to explore the impact of model improvements without retraining the model. The paper's focus on targeting the worst-off individuals in labor market settings is also a strength, as it highlights the potential for using predictive models to improve social outcomes. The authors' attempt to provide actionable insights for policymakers is commendable, as it demonstrates the practical relevance of their work. The paper also attempts to connect its work to existing literature on predictive modeling and fairness, which helps to contextualize its contributions. While the paper has several limitations, its attempt to address a complex problem with a novel framework is a significant strength.

✗ Weaknesses

After a thorough examination of the paper, I have identified several significant weaknesses that undermine its conclusions and limit its practical applicability. First and foremost, the paper's theoretical derivations rely heavily on the assumption of a bivariate normal distribution for the joint distribution of covariates (X) and outcomes (Y). This assumption, while common in some statistical literature, is not adequately justified within the context of this paper. The authors do not discuss the limitations of this assumption or the potential impact of deviations from normality on their results. Specifically, the paper fails to address how skewness or heavy tails in the distribution of X or Y could affect the accuracy of the derived policy value function and the PAR metric. Furthermore, the assumption of homoscedasticity, which is also implied by the bivariate normal distribution, is not discussed, and the paper does not consider the potential consequences of violating this assumption. This lack of sensitivity analysis is a major weakness, as real-world data often deviates from the idealized assumptions of normality and homoscedasticity. The absence of this analysis casts doubt on the robustness of the proposed framework and its applicability to real-world scenarios. My confidence in this weakness is high, as the paper explicitly states the bivariate normal assumption without further discussion of its limitations. Second, the paper's empirical validation is conducted solely on synthetic datasets. While the authors describe the synthetic data as mimicking real-world administrative data, the lack of detail in the data generation process makes it difficult to assess the realism of the experiments. The paper does not specify the distributions used for each variable, the sample size generation process, or how noise or bias is incorporated. This

lack of detail limits the generalizability of the findings, as the synthetic data may not fully capture the complexities and nuances of actual policy settings. The absence of real-world data validation is a significant weakness, as it raises concerns about the practical applicability of the proposed framework. My confidence in this weakness is high, as the paper explicitly states the use of synthetic data and lacks a detailed description of its generation process. Third, the paper does not provide a clear explanation of the motivation behind the specific formulation of the policy value function, $V(\alpha, \beta, R^2)$. While the components of the function are defined, the paper does not explain why this particular formulation, involving the bivariate normal CDF, is preferred over other potential functions. The choice of α , β , and R^2 as key parameters is presented as a given, without a detailed justification of their relevance and importance in this context. This lack of motivation makes it difficult to assess the novelty and significance of the proposed approach. My confidence in this weakness is high, as the paper defines the components of the policy value function but lacks a clear explanation of the rationale behind its specific formulation. Fourth, the paper introduces a residual scaling method to simulate improvements in prediction accuracy, but it does not provide a clear justification for the specific form of this method. The paper does not analyze the potential biases or inefficiencies introduced by this approximation, nor does it discuss the impact of this approximation on the Prediction-Access Ratio (PAR). This lack of analysis is a weakness, as it raises concerns about the validity of the results obtained using this method. My confidence in this weakness is high, as the paper mentions and applies residual scaling but lacks a detailed explanation, justification, and analysis of its properties and impact. Fifth, the paper lacks a thorough discussion of the limitations of the proposed framework. There is no discussion of the potential for the framework to be misused or to lead to unintended consequences. The assumptions underlying the framework are not clearly stated, and the conditions under which the framework is likely to be most effective are not discussed. The paper also does not address the sensitivity of the results to different choices of the screening threshold (α) and the targeted outcome quantile (β). The experiments use fixed values for these parameters without exploring how the results might change with different settings. This lack of sensitivity analysis is a weakness, as it limits the understanding of the robustness of the framework. My confidence in this weakness is high, as the paper lacks a dedicated discussion of limitations, potential misuse, and optimal use cases, and it does not perform a sensitivity analysis on α and β . Finally, the paper does not provide a detailed discussion of the computational complexity of the proposed framework. While the authors do not explicitly claim that the framework is computationally efficient, they do not provide a formal analysis of the time and space complexity of calculating the policy value function and the PAR metric. This lack of analysis is a weakness, as it limits the understanding of the scalability of the framework for large datasets. My confidence in this weakness is high, as the paper lacks any discussion or analysis of the computational complexity of the proposed framework.

Suggestions

To address the identified weaknesses, I recommend several concrete improvements. First, the authors should conduct a more thorough analysis of the sensitivity of their results to the assumption of bivariate normality. This should involve exploring alternative distributional assumptions, such as skewed or heavy-tailed distributions, and examining how the optimal trade-offs between prediction accuracy and screening access change under these different assumptions. The authors should also consider using robust statistical methods that are less sensitive to deviations from normality. Furthermore, they should explore the use of non-parametric methods that do not rely on distributional assumptions. This would help to establish the robustness of the proposed framework and its applicability to a wider range of real-world scenarios. Second, the authors should include experiments on real-world datasets to demonstrate the practical applicability of their framework. This would involve obtaining access to relevant data, pre-processing the data, and applying the proposed framework to this data. The experiments should be designed to evaluate the performance of the proposed method in realistic scenarios and to compare it with existing approaches. The authors should also provide a detailed description of the data used, including the sample size, the variables included, and the data collection process. This would allow other researchers to replicate their findings and assess the generalizability of the framework. If real-world data is not available, the authors should consider using more realistic simulation scenarios that incorporate features of real-world data, such as missing values, outliers, and measurement error. The authors should also discuss the limitations of using synthetic data and the potential impact of these limitations on the validity of their findings. Third, the authors should provide a more detailed justification for the specific formulation of the policy value function, $V(\alpha, \beta, R^2)$. This should include a discussion of alternative formulations and why they were not chosen. The authors should also provide a more intuitive explanation of what each parameter represents and how it relates to the policy evaluation problem. For example, they could explain how changes in α and β affect the trade-off between prediction accuracy and screening access, and how R^2 influences the overall policy value. This would make the paper more accessible to a broader audience and allow for a better understanding of the proposed method. Fourth, the authors should provide a more detailed justification for the proposed residual scaling method. This should include a theoretical analysis of the method and a discussion of its potential biases and inefficiencies. The authors should also compare the proposed method to other methods for improving predictive accuracy and discuss the trade-offs between these methods. Furthermore, they should analyze the impact of the residual scaling method on the Prediction-Access Ratio (PAR) and discuss the implications

of this impact. The authors should also consider using alternative methods for improving predictive accuracy, such as retraining the model or using ensemble methods. This would help to establish the effectiveness of the proposed method and its advantages over other methods. Fifth, the authors should include a more detailed discussion of the limitations of the proposed framework. This should include a discussion of the potential for the framework to be misused or to lead to unintended consequences. The authors should also acknowledge the assumptions underlying the framework and discuss the conditions under which the framework is likely to be most effective. Furthermore, they should conduct a sensitivity analysis of the results to different choices of the screening threshold (α) and the targeted outcome quantile (β). This analysis should explore how the policy value and PAR change as these parameters are varied, and it should provide practical guidance to policymakers on how to choose appropriate values for these parameters based on their specific policy goals and constraints. Finally, the authors should provide a detailed analysis of the computational complexity of their method and discuss its scalability for large datasets. This should include a formal analysis of the time and space complexity of calculating the policy value function and the PAR metric, as well as a discussion of potential strategies for improving computational efficiency.

 Questions

Based on my analysis, I have several questions that I believe are crucial for further understanding and improving the proposed framework. First, how sensitive are the results to deviations from the bivariate normality assumption? Have the authors considered alternative distributional assumptions, and how would these affect the derived policy value function and PAR metric? Specifically, what would be the impact of skewness or heavy tails in the distribution of X or Y, and how would the framework perform under heteroscedasticity? Second, can the authors provide empirical validation of their framework using real-world datasets? This would help to demonstrate the practical relevance and applicability of their proposed metrics. If real-world data is not available, can the authors provide more details about the synthetic data generation process, including the specific distributions and parameters used, and how they mimic real-world administrative data? Third, what is the computational complexity of the proposed framework, especially when dealing with large datasets? Are there any strategies for improving computational efficiency? A formal analysis of the time and space complexity of calculating the policy value function and the PAR metric would be beneficial. Fourth, how do the authors recommend choosing the screening threshold (α) and the targeted outcome quantile (β) in practice? Are there any guidelines or best practices for selecting these parameters based on specific policy contexts? A sensitivity analysis of the results to different choices of these parameters would be helpful. Fifth, can the authors provide more details about the residual scaling method? What is the theoretical justification for this method, and what are its potential biases and inefficiencies? How does this approximation impact the Prediction-Access Ratio (PAR) in operational settings? Finally, what are the limitations of the proposed framework, and what are the potential ethical considerations when applying it in real-world policy settings? A more thorough discussion of the potential for misuse and unintended consequences would be valuable.

 Scores

Soundness: 1.5

Presentation: 1.5

Contribution: 1.5

Rating: 2.5

 AI Review from ZGCA will be automatically processed

 Summary

The paper proposes a framework to quantify trade-offs between prediction accuracy and screening access in policy evaluation. It formalizes a policy value function $V(\alpha, \beta, R^2) = \Phi_z(z\alpha, z\beta; \rho)/\beta$ with $z\alpha = \Phi^{-1}(\alpha)$, $z\beta = \Phi^{-1}(\beta)$, and $\rho = \sqrt{R^2}$, and introduces the Prediction-Access Ratio (PAR) to compare finite gains in policy value from increasing screening access ($\Delta\alpha$) versus improving predictive performance (ΔR^2). The authors derive local sensitivities (e.g., $\partial V/\partial\alpha \approx 1.77513$, $\partial V/\partial R^2 \approx 0.61282$), and conduct synthetic experiments showing that residual scaling ($\delta = 0.1$) increases test R^2 from 0.16866 to 0.32661 and empirical $V(\alpha, \beta)$ from 0.70 to 0.80, while a modest increase in screening threshold ($\Delta\alpha \approx 0.03$) can yield comparable gains. They argue the framework informs whether to invest in model accuracy or screening capacity under resource constraints.

 Strengths

- Conceptual clarity in formalizing a policy value function that jointly depends on screening threshold, outcome quantile, and predictive fit (Section 1; Equation defining $V(\alpha, \beta, R^2)$).
- Introduction of PAR as a simple, interpretable comparative metric for finite trade-offs between screening access and predictive accuracy (Section 4).
- Sensitivity analysis with explicit local derivatives ($\partial V/\partial\alpha$ and $\partial V/\partial R^2$) offering actionable intuition for policy tuning (Sections 1 and 6).
- Capacity gap analysis highlighting that small increases in screening access can rival gains from predictive model improvements ($\Delta\alpha^* \approx 0.03$; Sections 1, 6, 7).
- Practical motivation for resource-constrained agencies and a discussion connecting to fairness and policy deployment (Sections 1, 7).

 Weaknesses

- Heavy reliance on a bivariate normal approximation and a non-justified mapping $\rho = \sqrt{R^2}$. The link between out-of-sample R^2 of a predictive model and the correlation parameter of the joint $(\tilde{Y}, Y)$ distribution is not derived or validated; this is central to the framework's interpretability and may not hold under misspecification, nonlinearity, or heteroscedasticity (Sections 1–2, 4).
- Empirical evaluation uses a very small synthetic dataset (Train 169 / Val 69 / Test 62) and reports single point estimates only. No confidence intervals, no multiple seeds, no variance estimates, and no stress tests are provided (Section 5–6), making claims about ΔV or PAR fragile.
- Residual scaling ($\delta = 0.1$) as a proxy for prediction improvement is under-specified and risks being a post-hoc manipulation that may not reflect feasible or causal model improvements. Its statistical validity and operational meaning are unclear (Sections 5–7).
- Related work and positioning are incomplete and sometimes tangential (e.g., LTE access reservation, spectrum sharing) relative to the ML policy evaluation and off-policy evaluation literature; connections to causal uplift/risk ranking, cost-sensitive learning, or selective prediction are limited (Section 3).

- Theoretical development remains partial: key properties (e.g., monotonicity and convexity of V in α and ρ , conditions under which PAR is stable across β) are stated empirically rather than proved; derivations for sensitivities and the ρ - R^2 mapping are not provided (Sections 4, 6).
- No real-world dataset evaluation, no ablations across non-normal distributions or heteroscedastic noise, and no subgroup analyses beyond a brief gender note with identical V values (Section 6) limit generalizability and fairness insights.
- Reproducibility details are incomplete for the simpler models and simulations (e.g., comprehensive seeding, code release).

? Questions

- Please justify the mapping $\rho = \sqrt{R^2}$. Under what assumptions on $(\tilde{Y}, Y)$ and the predictive model does this equality hold? How is ρ estimated in practice from out-of-sample predictions, and how sensitive are conclusions to estimation error?
- Can you provide formal derivations for $\partial V / \partial \alpha$ and $\partial V / \partial R^2$, and any monotonicity/convexity properties of V with respect to α , β , and ρ ? Under what conditions does PAR remain stable across operating points?
- What exactly is the residual scaling procedure ($\delta = 0.1$)? Does it use only training data, or any test-time information? How does it avoid leakage, and how would one achieve similar gains through retraining or calibration in practice?
- How were $\alpha = 0.2$ and $\beta = 0.15$ chosen? Have you explored sensitivity of PAR and $\Delta \alpha^*$ to these operating points, including cost constraints or prevalence shifts?
- Can you report results with uncertainty (e.g., bootstrap CIs) and multiple random seeds, and on larger or real datasets to demonstrate robustness and generalizability?
- How does the framework behave under non-normal and heteroscedastic settings (e.g., heavy tails, mixture distributions)? Can you provide simulations that relax the bivariate normal assumption?
- How does PAR integrate with costs (screening budget, false positive/negative costs)? Could you provide a cost-weighted decision rule that translates PAR into actionable resource allocation?
- What is the relationship between V and standard ranking metrics (AUC, precision@k) when β and α are small? Can you connect V to known metrics to aid interpretability?

! Limitations

- Strong distributional assumptions (bivariate normality, homoscedasticity) may fail in real administrative data, potentially biasing V and PAR estimates.
- Small synthetic datasets and point estimates limit statistical reliability; results may not generalize without larger-scale or real-data validation.
- Residual scaling as an approximation to improved prediction lacks theoretical grounding and may not reflect deployable enhancements.
- Potential negative societal impacts include inequitable resource allocation if α is tuned without subgroup-specific constraints, and burdening limited screening resources if $\Delta \alpha$ is increased without capacity planning.
- Framework currently omits explicit cost modeling; decisions based solely on V or PAR could misalign with budgetary or ethical constraints.
- Limited fairness analysis beyond gender; untested across sensitive attributes or distribution shift could mask disparate impacts.

Image Evaluation

Cross-Modal Consistency: 34/50

Textual Logical Soundness: 22/30

Visual Aesthetics & Clarity: 16/20

Overall Score: 72/100

Detailed Evaluation (≤ 500 words):

Image-first understanding

- Figure 1(a): Screening policy curve. Axes: $x=Y$, $y=\Pr[\hat{y} \leq \hat{q}\alpha \mid Y]$. Smoothly decreasing from ≈ 1 (low Y) to ≈ 0 (high Y).
- Figure 1(b): Heatmap of ΔV from $\Delta R2=0.01$ over $(\alpha, R2)$. Axes: $x=\alpha$ (0.05–0.5), $y=R2$ (≈ 0 –0.9), colorbar up to ≈ 0.025 ; higher ΔV at low $R2$ and mid α .
- Synopsis: (a) shows conditional selection behavior at fixed parameters; (b) maps global sensitivity of V to small $R2$ changes across α and $R2$.

1. Cross-Modal Consistency

- Major 1: Claim “random screening ($R2=0$) $\Rightarrow V=0$ ” contradicts the stated $V(\alpha, \beta, R2)$ formula (which implies $V=\alpha$ when $p=0$). Evidence: Sec 6, “random screening (simulating $R2=0$) results in $V = 0.00000$ ”.
- Major 2: Central “capacity gap” result ($\Delta\alpha^*\approx 0.0300$) is repeatedly asserted without a dedicated figure/table or precise setup (α, β , baseline V , uncertainty). Evidence: Sec 1, “ $\Delta\alpha^* \approx 0.0300$ can yield gains comparable...”, and Sec 6 same claim.
- Minor 1: Fig. 1(a) caption specifies “ $\alpha=0.2$ & $R2=0.5$ ” but the plot lacks these annotations; readers cannot verify parameterization from the graphic. Evidence: Fig. 1 caption.
- Minor 2: PAR differs across sections ($\approx 2.83/2.32$ in theory vs 2.00 empirically) without explicit alignment of conditions ($\alpha, \beta, R2$, finite steps). Evidence: Sec 6 vs Sec 4 tables.
- Minor 3: Results note DT $R2=0.20332$ vs complex 0.16866 (before scaling), which could confuse expectations about “complex vs simple.” Evidence: Sec 6, “Decision Tree...0.20332” and “complex model...0.16866”.

2. Text Logic

- Major 1: Theoretical inconsistency on $R2=0$ undermines the framework’s correctness and downstream claims relying on V ’s baseline. Evidence: Sec 6, “ $V = 0.00000$ ”.
- Minor 1: Excessive precision/spacing in thresholds impedes readability. Evidence: Sec 5, “Q u a n t i l e _ { 0 . 2 } (\hat{y}) \approx -0.8314...”.
- Minor 2: Some related-work items (LTE/cognitive radio) are referenced but weakly tied to the core framework’s implications.

3. Figure Quality

- Minor 1: Fig. 1(a) lacks $\alpha, \beta, R2$ and p in legend/annotation; add values and brief equation snippet for figure-alone comprehension.
- Minor 2: Fig. 1(b) colorbar/font sizes are small; add numeric tick labels and mark the evaluation point (e.g., $\alpha=0.2$, $R2=0.5$).

Key strengths:

- Clear formalization of $V(\alpha, \beta, R2)$ and operational PAR metric.
- Useful sensitivity visualization (ΔV heatmap) and concrete empirical tables.
- Practical framing of screening-vs-prediction trade-offs.

Key weaknesses:

- The $R2=0 \Rightarrow V=0$ claim contradicts the model (should yield $V=\alpha$); fix and reconcile results.
- Capacity-gap $\Delta\alpha^*$ lacks concrete evidence (figure/table, CI, setup).
- Parameter annotations missing in Fig. 1(a); precision/formatting issues reduce clarity.

 Scores

Originality: 3

Quality: 2

Clarity: 2

Significance: 2

Soundness: 2

Presentation: 2

Contribution: 2

Rating: 4

 AI Review from SafeReviewer will be automatically processed Summary

This paper introduces a framework for evaluating policy interventions by quantifying the trade-off between prediction accuracy and screening access. The authors propose a policy value function, $V(\alpha, \beta, R^2)$, which integrates the screening threshold (α), the targeted outcome quantile (β), and the predictive performance (R^2). A key contribution is the Prediction-Access Ratio (PAR), which aims to quantify the relative benefits of improving prediction accuracy versus expanding screening access. The authors employ a bivariate normal approximation to derive the policy value function and conduct simulation experiments on synthetic datasets to validate their framework. The core idea revolves around balancing the accuracy of predictive models with the reach of policy interventions, recognizing that both factors influence the overall effectiveness of policy implementation. The authors argue that even modest improvements in either prediction accuracy or screening access can lead to significant gains in policy outcomes, and the PAR metric is designed to help policymakers determine where to focus their efforts. The methodology involves generating synthetic data, training predictive models, and then evaluating the policy value function under different scenarios. The simulation experiments compare complex models (Gradient Boosting and CatBoost) with simpler models (Decision Trees) and explore the impact of

residual scaling on predictive performance. The main empirical finding is that a small increase in the screening threshold can yield gains comparable to those achieved by improving the prediction model, suggesting a potential trade-off between these two aspects of policy design. The authors also conduct a capacity gap analysis to quantify the minimal additional screening threshold required for simpler models to match the performance of more complex models. While the paper presents a novel framework for analyzing this trade-off, the analysis is limited by the use of synthetic data and the lack of real-world validation. The paper's significance lies in its attempt to provide a quantitative approach to balancing prediction accuracy and screening access, which is a critical consideration for many policy interventions. However, the practical applicability of the framework remains unclear without empirical validation on real-world datasets. The paper also lacks a detailed discussion of the cost implications of expanding screening access versus improving prediction accuracy, which is a crucial factor for policymakers. Despite these limitations, the paper offers a valuable starting point for further research in this area.

✓ Strengths

I find the paper's core strength lies in its attempt to formalize the often-overlooked trade-off between prediction accuracy and screening access in policy evaluation. The introduction of the policy value function, $V(\alpha, \beta, R^2)$, is a novel contribution that provides a unified framework for considering both aspects of policy design. This function, derived from a bivariate normal approximation, allows for a quantitative assessment of policy outcomes based on the screening threshold, the targeted outcome quantile, and the predictive performance of the model. The Prediction-Access Ratio (PAR), while not fully justified in its current form, is a valuable attempt to quantify the relative benefits of improving prediction accuracy versus expanding screening access. This metric has the potential to guide policymakers in allocating resources effectively, by providing a way to compare the impact of different interventions. The paper's use of simulation experiments, although limited to synthetic data, provides a controlled environment for exploring the behavior of the proposed framework. The comparison of complex models (Gradient Boosting and CatBoost) with simpler models (Decision Trees) is a practical approach that highlights the potential for achieving comparable policy outcomes with simpler models and expanded screening access. The capacity gap analysis, which quantifies the minimal additional screening threshold required for simpler models to match the performance of more complex models, is another practical contribution that can inform policy decisions. The paper's focus on the practical implications of the trade-off between prediction accuracy and screening access is also a strength. The authors recognize that policy interventions are often constrained by limited resources, and that there is a need to balance the desire for accurate predictions with the need to reach a larger proportion of the population. The paper's attempt to provide a quantitative framework for addressing this challenge is a valuable step towards more effective policy design. The paper also attempts to bridge the gap between theoretical constructs and practical policy applications, which is a crucial step for translating academic research into real-world impact. The authors' recognition of the need for transparent and quantitatively validated decision protocols is also a positive aspect of the paper.

✗ Weaknesses

After a thorough examination of the paper, I've identified several significant weaknesses that warrant careful consideration. First, the paper suffers from a lack of clarity in its writing and presentation. The frequent use of numerical examples, while intended to illustrate the concepts, often disrupts the flow of the text and makes it difficult to grasp the underlying ideas. For instance, the abstract, introduction, and background sections are filled with specific numerical values, such as "Test R^2 improves from 0.16866 to 0.32661" and " $V(\alpha, \beta)$ increases from 0.70000 to 0.80000", which do not contribute to a clear understanding of the methodology and instead make the text cumbersome. This excessive use of numerical examples, especially before the results section, makes the paper less readable and obscures the core concepts. Second, the paper's core contribution, the Prediction-Access Ratio (PAR), lacks a strong theoretical justification. While the authors introduce PAR as a metric to quantify the relative impact of finite improvements in screening thresholds versus enhancements in predictive accuracy, they do not provide a clear explanation of why this specific ratio is the most appropriate measure. The paper does not compare PAR to other potential metrics or provide a detailed analysis of its properties and limitations. The definition of PAR as $\Delta V / \Delta R^2 / \Delta V / \Delta \alpha$ is presented without a deep dive into its theoretical underpinnings, and the paper does not explore alternative definitions or justify why this particular formulation is optimal. This lack of theoretical grounding weakens the credibility of the proposed metric. Third, the paper's reliance on synthetic data is a major limitation. While the authors mention that the synthetic data is designed to mimic real-world administrative data, they do not provide sufficient details about the data

generation process or justify why synthetic data is sufficient for validating their framework. The paper lacks a detailed description of the distributions used to generate the data, the parameters of these distributions, and the rationale behind these choices. This lack of transparency makes it difficult to assess the validity of the simulation results and their generalizability to real-world scenarios. The absence of real-world data experiments significantly limits the practical applicability of the proposed framework. Fourth, the paper lacks a discussion of the cost implications of expanding screening access versus improving prediction accuracy. The paper focuses on the quantitative aspects of the trade-off but does not consider the cost of increasing the screening threshold, which may involve additional resources, personnel, or infrastructure. Similarly, the paper does not discuss the cost of improving prediction accuracy, such as the cost of collecting more data or developing more complex models. This omission is a significant weakness, as cost is a crucial factor for policymakers when making decisions. Fifth, the paper's methodological justification for using residual scaling to simulate improved prediction accuracy is weak. The authors do not provide a strong theoretical basis for this approach, nor do they discuss its limitations or potential biases. The paper does not explore alternative methods for simulating improved prediction accuracy, such as using ensemble methods or more sophisticated modeling techniques. This lack of methodological rigor raises concerns about the validity of the simulation results. Sixth, the paper's literature review is not comprehensive. The paper does not adequately discuss the existing literature on screening and diagnostic testing, which has extensively studied the trade-off between the sensitivity and specificity of tests, which is directly related to the paper's topic. The paper also does not discuss the literature on cost-effectiveness analysis in healthcare, which is relevant to the paper's focus on balancing benefits and costs. This lack of engagement with relevant literature weakens the paper's contribution and its connection to existing knowledge. Finally, the paper's experimental setup lacks sufficient detail. While the authors provide some information about the dataset and the models used, they do not provide enough detail to allow for reproducibility. For example, the paper does not specify the exact procedure for generating the synthetic data, the specific parameters used for the models, or the details of the experimental protocol. This lack of detail makes it difficult to assess the validity of the results and their generalizability to other settings. The paper also lacks a thorough discussion of the limitations of the proposed framework and potential avenues for future research. These weaknesses, taken together, significantly limit the paper's impact and its practical applicability.

Suggestions

To address the identified weaknesses, I recommend several concrete improvements. First, the authors should significantly revise the paper's writing style to improve clarity and readability. This involves removing the excessive use of numerical examples from the abstract, introduction, and background sections, and focusing on explaining the core concepts in a clear and concise manner. The authors should use more general terms and avoid getting bogged down in specific numerical values until the results section. Second, the authors should provide a more rigorous theoretical justification for the Prediction-Access Ratio (PAR). This involves exploring alternative definitions of the metric, comparing it to existing metrics, and providing a detailed analysis of its properties and limitations. The authors should also discuss the assumptions underlying the metric and the conditions under which it is most effective. This would strengthen the theoretical foundation of the proposed metric and increase its credibility. Third, the authors must include experiments on real-world datasets to validate their framework. This involves obtaining and using publicly available datasets that are relevant to the policy problem being addressed. The authors should also provide a detailed description of the data, including the variables used, the sample size, and any preprocessing steps. This would significantly increase the practical applicability of the proposed framework and demonstrate its effectiveness in real-world scenarios. Fourth, the authors should incorporate a cost-benefit analysis into their framework. This involves quantifying the cost of expanding screening access and the cost of improving prediction accuracy. The authors should also consider the potential costs associated with false positives and false negatives. This would provide a more comprehensive understanding of the trade-offs involved and allow for more informed decision-making. Fifth, the authors should provide a stronger methodological justification for using residual scaling to simulate improved prediction accuracy. This involves exploring alternative methods for simulating improved prediction accuracy and discussing the limitations and potential biases of residual scaling. The authors should also provide a theoretical basis for this approach and explain why it is a valid way to simulate improved prediction accuracy. Sixth, the authors should expand their literature review to include relevant work on screening and diagnostic testing, as well as cost-effectiveness analysis in healthcare. This would help to contextualize their work and demonstrate its contribution to the field. The authors should also discuss how their approach differs from existing methods and what advantages it offers. Seventh, the authors should provide more detail about their experimental setup, including the exact procedure for generating the synthetic data, the specific parameters used for the models, and the details of the experimental protocol. This would improve the reproducibility of the results and allow for a more thorough evaluation of the proposed framework. Finally, the authors should include a more thorough discussion of the limitations of their framework and potential avenues for future research.

This would help to contextualize the findings and provide a roadmap for future work in this area. These suggestions, if implemented, would significantly improve the quality and impact of the paper.

? Questions

After reviewing the paper, I have several questions that I believe are critical for further understanding and development of the proposed framework. First, I am curious about the specific assumptions underlying the bivariate normal approximation used to derive the policy value function. How sensitive are the results to violations of these assumptions, and what alternative distributional assumptions could be considered? This is important because the validity of the policy value function relies on the accuracy of this approximation. Second, I would like to understand the rationale behind the specific definition of the Prediction-Access Ratio (PAR). Why is the ratio of the relative change in policy value due to changes in R^2 to the relative change in policy value due to changes in α the most appropriate metric for quantifying the trade-off? Are there other potential metrics that could be considered, and what are the advantages and disadvantages of each? This is crucial for understanding the theoretical basis of the proposed metric. Third, I am interested in the details of the synthetic data generation process. What specific distributions were used to generate the data, and what parameters were used for these distributions? How were the relationships between the variables defined, and what assumptions were made about these relationships? This is essential for assessing the validity and generalizability of the simulation results. Fourth, I would like to know more about the cost implications of expanding screening access versus improving prediction accuracy. How can these costs be quantified, and how can they be incorporated into the proposed framework? What are the potential trade-offs between the cost of expanding screening access and the cost of improving prediction accuracy? This is critical for making informed policy decisions. Fifth, I am curious about the limitations of using residual scaling to simulate improved prediction accuracy. What are the potential biases introduced by this approach, and what alternative methods could be used to simulate improved prediction accuracy? How does the choice of the scaling factor δ affect the results? This is important for understanding the validity of the simulation experiments. Sixth, I would like to understand how the proposed framework can be applied to different policy contexts. What are the key considerations for adapting the framework to different settings, and what are the potential challenges involved? This is crucial for assessing the practical applicability of the framework. Finally, I am interested in the potential for incorporating fairness considerations into the proposed framework. How can the framework be modified to ensure that policy interventions are not only effective but also equitable? This is important for ensuring that the framework is used in a responsible and ethical manner. These questions are intended to probe the core methodological choices and assumptions of the paper, and I believe that addressing them would significantly strengthen the paper's contribution.

Scores

Soundness: **1.75**

Presentation: **1.5**

Contribution: **1.75**

Rating: 3.5

Keywords

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